

House Price Prediction — Project Summary

Objective

In this project, we have investigated a **dataset related to housing (545 listings, 13 features)** in order to determine which characteristics are significant in predicting the selling price.

Methodology

- **Cleaned the data:** checked for missing values and duplicates (found none), and one-hot encoded categorical fields (e.g. main road, furnishing status) into numeric form.
- **Split the data** 80/20 into training and test sets.
- **Trained and compared two models:** Linear Regression and Random Forest Regressor, evaluated using MAE, RMSE, and R^2 .

Model Performance

Model	MAE	RMSE	R^2
Linear Regression	₹970,043	₹1,324,507	0.653
Random Forest	₹1,022,560	₹1,401,497	0.611

Linear Regression slightly outperformed Random Forest on every metric, on this dataset and split.

Key Insights

- **Biggest price drivers:** Area and number of bathrooms showed the strongest positive correlation with price, followed by air conditioning, number of stories, and parking. Being unfurnished was the strongest negative factor.
- **Model accuracy in plain terms:** The Linear Regression model accounts for around 65% of price variance ($R^2 = 0.653$) with an average error of approximately ₹970,000, which amounts to 20% of the average housing price (₹4.77M). It can be useful as a rough estimate but cannot be used to determine an accurate value.
- **What was surprising:** The Linear Regression model performed better than Random Forest, indicating that the prices increased in a linear fashion based on variables such as area and bathroom count in the particular dataset size (545 observations), not on any complicated interactions. The distribution was also positively skewed, implying that the mean price exceeded the median price because of high-value houses.
- **Recommendation for a real estate business:** prioritize and market larger area, additional bathrooms, and air conditioning, as these have the most leverage on perceived value, while features like hot water heating have minimal influence on price and are not worth emphasizing in listings.